Inorganic Cumulative Exam
September 6 2024

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology, level of detail, and examples as answers to these questions. For some of these, the goal is to determine how well you can apply knowledge to a question with an indeterminant or qualitative answer.

1. (a) (11 pts) Draw the molecular orbital diagram of iodine monofluoride (IF) and comment on <u>the bond order</u> <u>and any molecular polarity that exists</u>. Be explicit in any assumptions that you make. For reference, the orbital energies of interest to this problem are (in eV): F (2s), -40.17; F (2p), -18.65; I (5s), -17.9; I (5p), -8.2;
   (b) (9 pts, 3 pts each) Sketch the bonding orbital, the highest occupied molecular orbital, and the lowest unoccupied molecular orbital for IF.

2. Draw each of the following molecules and assign the corresponding point group (5 points each): CCl_4, CCl_3F, CCl_2F_2, HClBrF. *Note: you do not need character tables to do this.*

3. (a) (10 pts) What is the difference between point symmetry and space symmetry?
   (b) (6 pts) Give three examples of a chemical compound with point symmetry and identify the point group of each.
   (c) (4 pts) Provide an example of a chemical compound with space symmetry and identify the primitive Bravais lattice and basis for the compound.

4. Consider three ions of Co^{3+}: $[Co(H_2O)_6]^{3+}$, $[Co(NH_3)_6]^{3+}$, and $[CoF_6]^{3-}$.
   (a) (5 pts) Draw the octahedral field splitting diagram for the generic Co^{3+} ion, including valence electrons.
   (b) (5 pts) In aqueous solution, the ions appear to have different colors. Explain the physical origin of their color, including interactions with light (assume we are discussing what these solutions look like under the typical overhead white lights of the chemistry lab).
   (c) (10 pts) In aqueous solutions, one ion is orange, one is green, and one is blue. Match the apparent color to the ion and explain your rationale.

5. (a) (5 pts) Write the Schrödinger equation for a one-dimensional electron in a general potential, *V(x)*.
   (b) (15 pts) How does the energy of solutions to the Schrödinger equation change with the number of nodes in the solution? Use the form of the Schrödinger equation to justify your answer. *Hint*: *the first term of the SE is important here.*

6. While the band gap of SnO_2 and TiO_2 (both greater than 3.0 eV) are larger than for Fe_2O_3 (roughly 2.4 eV), minority carriers (holes in all cases) are more mobile through SnO_2 and TiO_2 than for Fe_2O_3. Compare the mechanisms of conductivity in each compound and describe the issues that limit hole conductivity in Fe_2O_3.

7. LiBr has the rock salt crystal structure (NaCl) and a density of 3.46 g cm^{-3}.
   (a) (5 pts) Draw the LiBr unit cell.
   (b) (15 pts) What is the interionic distance between Li and Br ions? Explain any assumptions you make.

8. (a) (10 pts) Describe the radius-ratio rule for the structure of binary ionic compounds.
   (b) (10 pts) Do large cations in ionic compounds generally have a large or small coordination number? Explain your answer, including qualitatively describing what 'large' or 'small' coordination means in your explanation.

9. Silicon nanoparticles can be synthesized by heating silica in the presence of metallic Mg.
   (a) (10 pts) Describe the Stöber reaction for synthesizing SiO_2 from tetraethylorthosilicate. What other compounds are necessary?
   (b) (10 pts) Describe the reaction between Mg (s) and SiO_2(s) that leads to nanoparticulate, elemental Si. Comment on whether this reaction is endothermic and provide a chemical rationale for your answer.

10. (a) (13 pts) Describe x-ray absorption spectroscopy (XAS) as a materials characterization technique. Include a description of the measurements process, necessary equipment, and an example experiment that uses XAS to measure some property of a compound.
    (b) (7 pts) How are the more specialized XAS techniques (1) extended x-ray absorption fine structure (EXAFS) and (2) x-ray absorption near-edge spectroscopy (XANES) different? Use examples such as their corresponding spectral region, information from measurements, and data processing needs.

PERIODIC TABLE OF ELEMENTS

Atomic # / Symbol / Name / Weight

- C Solid
- Hg Liquid
- H Gas
- Rf Unknown

Metals: Alkali metals; Alkaline earth metals; Lanthanoids (Lanthanides); Actinoids (Actinides); Transition metals; Post-transition metals

Metalloids

Nonmetals: Other nonmetals; Noble gases

Pnictogens; Chalcogens; Halogens

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

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